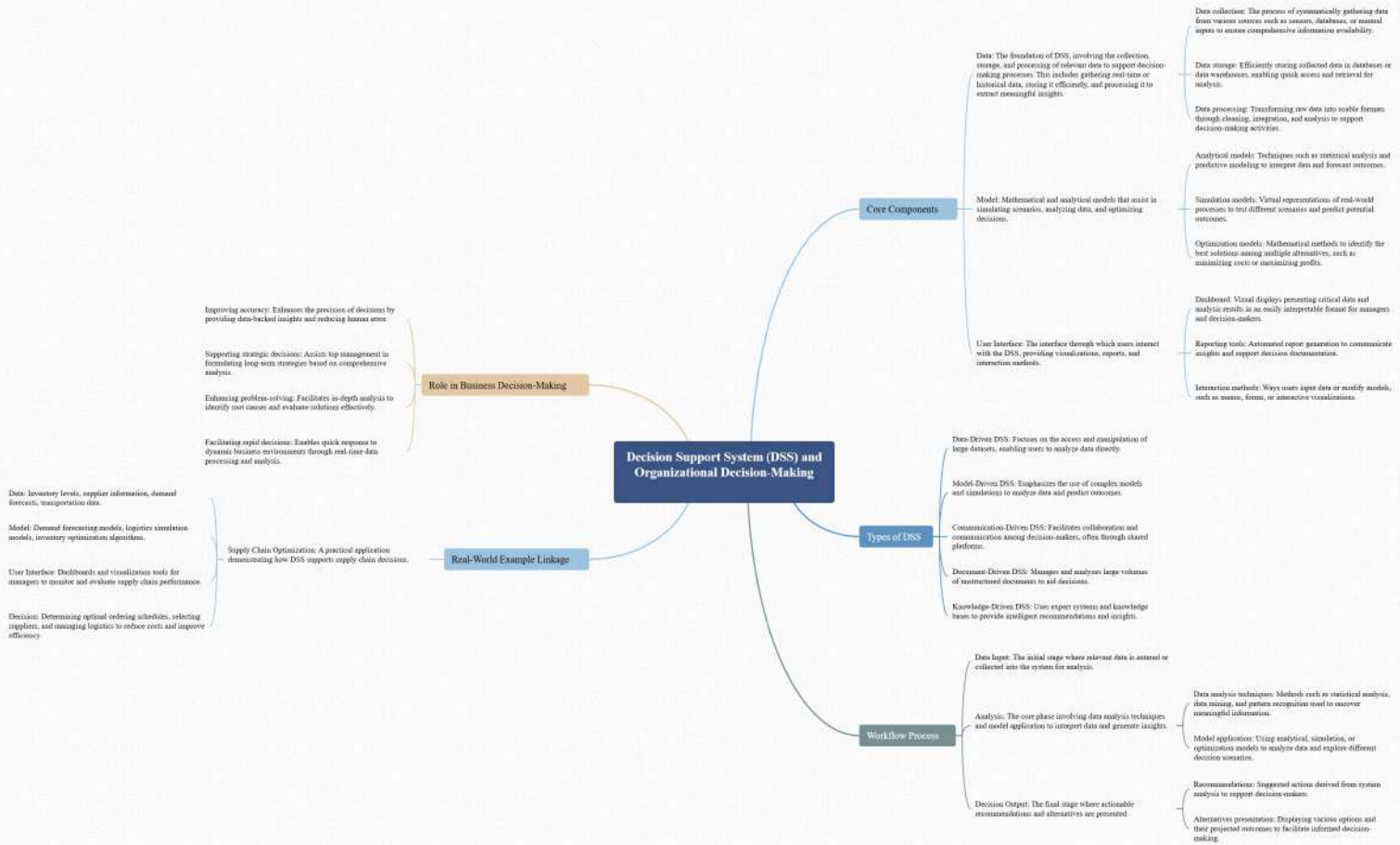
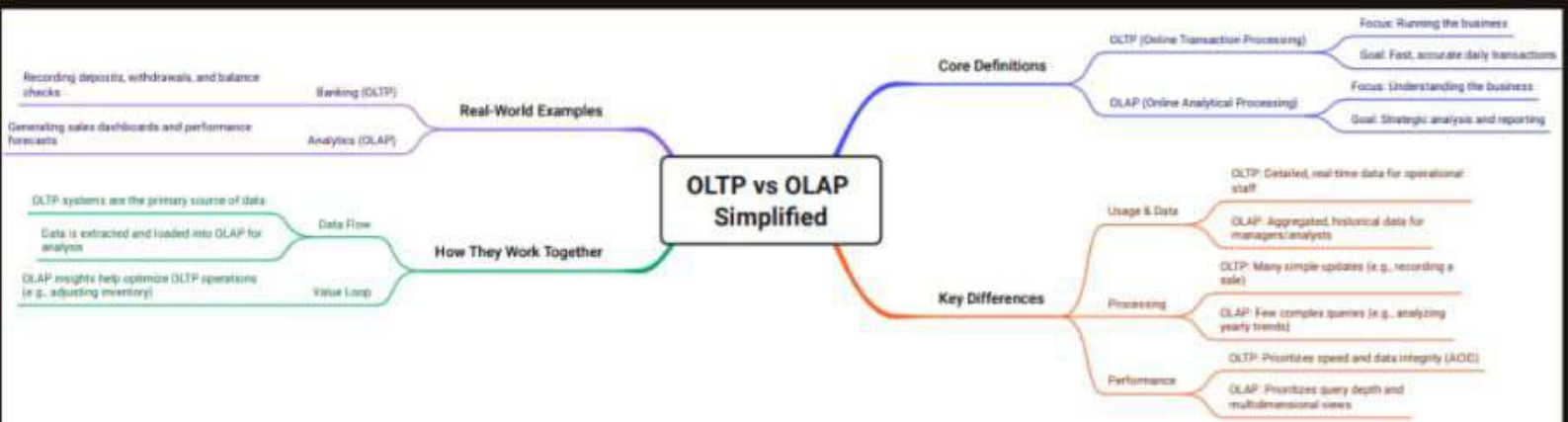






Architecture of a Data Warehouse

Layers · Data Flow · Component Roles

Data Flow Overview (Source → End User)



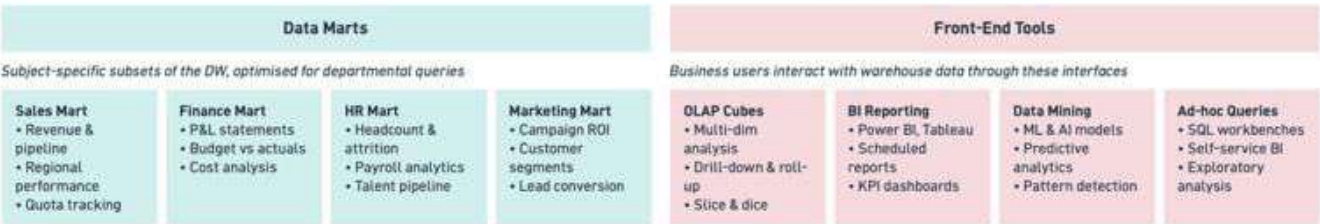
Layer Details — Sources → Warehouse



Metadata Layer — The Knowledge Backbone



Data Marts & Front-End Tools



Role of Each Layer — Summary

Layer	Primary Role	Data State
Data Sources	Originates raw, live transactional data from operational systems	Raw / Live
ETL Layer	Extracts, transforms and loads data into the warehouse	In-flight
Staging Area	Temporary buffer for raw data before transformation	Unvalidated
Central Data Warehouse	Centralised, integrated, historical subject-oriented store	Integrated
Metadata Layer	Describes structure, origin and meaning of all warehouse data	Descriptive
Data Marts	Subject-specific subsets for departmental analytical use	Aggregated
Front-End Tools	Business users access data via reports, OLAP and dashboards	Presented

QUERY PROCESSING INTEGRATED WITH BI TOOLS: FOCUS ON KNIME

1. QUERY PROCESSING STAGES

Converts a user query into results through systematic steps.

- 1. PARSING**
 - Check syntax & semantics
 - Validate objects, tables, columns
 - Generate parse tree / logical plan
- 2. OPTIMIZATION**
 - Apply rewrite rules
 - Choose efficient join order
 - Select access paths
 - Cost-based optimization
 - Generate optimized plan
- 3. EXECUTION**
 - Execute the optimized plan
 - Access data (scan, join, aggregate, sort, etc.)
 - Produce result set

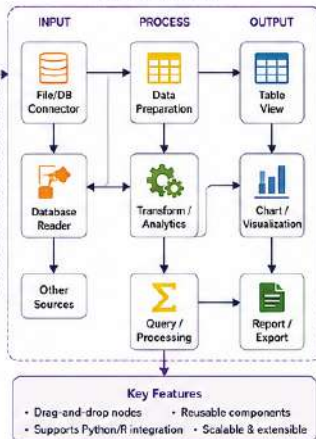
2. ROLE OF BI TOOLS

Bridge between technical processing and business users.

- Connect to multiple data sources (DB, files, DW, cloud, APIs)
- Provide self-service analytics for business users
- Transform complex data into meaningful insights
- Ensure data governance, security and reuse of workflows
- Enable reporting, dashboards and decision support

3. KNIME WORKFLOW STRUCTURE

Visual, modular and extensible analytics platform.



4. DATA → QUERY → VISUALIZATION PIPELINE

End-to-end flow from raw data to actionable insights.

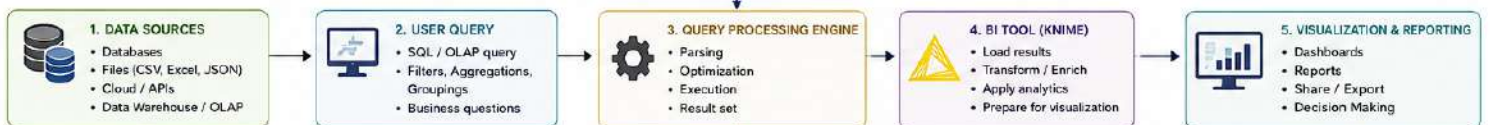
- DATA**
 - Structured/Unstructured data
 - Multiple sources
 - Data warehouse / Data lake
- QUERY**
 - User query (OLAP/SQL)
 - Processed by query engine (Parsing → Optimization → Execution)
 - Return result set
- VISUALIZATION**
 - Convert results into visuals
 - Dashboards, charts, tables
 - Interactive exploration
 - Share & report

5. OLAP APPLICATION INTEGRATION

KNIME integrates OLAP capabilities for multidimensional analysis.

- OLAP DATA SOURCE**
 - Data warehouse
 - OLAP server (Mondrian, Microsoft SSAS, SAP BW)
 - Multidimensional models
- OLAP OPERATIONS**
 - Slice
 - Dice
 - Drill-down
 - Roll-up
 - Pivot (Rotate)
- INTEGRATION IN KNIME**
 - Use dedicated nodes to connect, query and analyze OLAP cubes.
- Examples of KNIME Nodes**
 - OLAP Connector
 - OLAP Cube View
 - OLAP Dimension View

HOW IT ALL CONNECTS



OUTCOME / BENEFITS

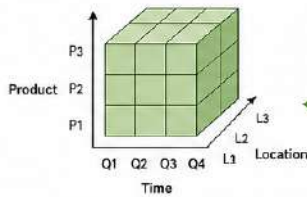
-  Faster insights through optimized query processing
-  Multidimensional analysis with OLAP inside KNIME
-  Self-service analytics for business users
-  Consistent, governed and reusable workflows
-  Better decisions with interactive reports & dashboards

OLAP OPERATIONS, DATA CUBES & DIMENSIONAL MODELLING

1. DATA CUBE STRUCTURE (Dimensions, Measures)

DATA CUBE

A multi-dimensional structure that organizes data for fast analytical processing.



DIMENSIONS

- Perspectives by which data is analyzed.
- Examples:
 - Time
 - Product
 - Location
 - Customer

MEASURES

- Quantitative data for analysis.
- Examples:
 - Sales Amount
 - Quantity Sold
 - Profit
 - Count of Orders

2. OLAP OPERATIONS

Operations performed on data cubes to view data from different angles.

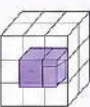
SLICE

Fix one dimension to get a sub-cube.
Example: Sales in 2024



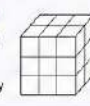
DICE

Select a sub-cube by choosing a subset of values in multiple dimensions.
Example: Sales of Product P1 & P2 in Q1 & Q2



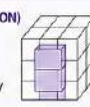
ROLL-UP (AGGREGATION)

Summarize data by climbing up a dimension hierarchy.
Example: Daily → Monthly → Yearly



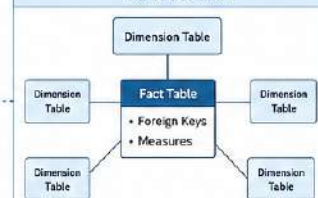
DRILL-DOWN (DISAGGREGATION)

Go from summarized data to more detailed data.
Example: Yearly → Monthly → Daily



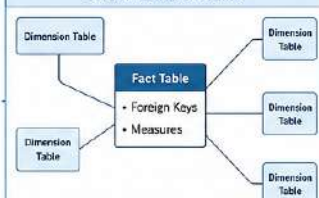
3. DIMENSIONAL MODELLING TECHNIQUES

STAR SCHEMA



- Central fact table connected to denormalized dimension tables.
- Simple, fast queries.
- Best for performance.

SNOWFLAKE SCHEMA



- Dimension tables normalized into multiple related tables.
- Saves storage.
- More complex queries.

4. FACT AND DIMENSION TABLES

FACT TABLE

- Stores quantitative data (measures).
- Contains foreign keys to dimensions.
- Grain: the level of detail.
- Example: Sales_Fact

Date_ID	Product_ID	Store_ID	Sales	Quantity	Profit
1	101	10	1000	5	200
1	102	10	800	4	120
2	101	11	1200	6	300

DIMENSION TABLE

- Stores descriptive attributes.
- Provides context to facts.
- Example: Product_Dim

Product_ID	Product_Name	Category	Brand
101	Smartphone	Electronics	BrandA
102	Laptop	Electronics	BrandB
103	Tablet	Electronics	BrandC

5. ANALYTICAL USAGE



BUSINESS ANALYSIS

Analyze trends, patterns and performance.



PERFORMANCE MONITORING

Track KPIs across different dimensions.



DECISION SUPPORT

Quick insights for strategic decisions.



REPORTING & DASHBOARDS

Power BI, Tableau, etc. for visualization.



PLANNING & FORECASTING

Use historical analysis for future planning.